UNIVERSITY OF EASTERN AFRICA, BARATON

INSY 281: SYSTEM ANALYSIS & DESIGN

BASELINE PROJECT PLAN (BPP)

FarmLink: Integrated Farm Produce Marketplace and Management Information System

Prepared by: Thabita Jeptoo
Course: INSY 281 – System Analysis & Design
Prepared for: Mr. Jefferson Mwatati
Academic Year: 2026
Submission Deadline: 4 November 2026

1. PROJECT OVERVIEW

FarmLink is a proposed web-based agricultural information system designed to connect farmers directly with buyers while improving product visibility, order management, communication, and administrative decision-making. The system will provide a centralized platform where registered farmers can create profiles, list farm produce with photographs, prices, quantities and availability, while buyers can search, compare, order and communicate with sellers. Administrators will monitor users, products, transactions, reports and system activities.

The project addresses common challenges in agricultural marketing, including dependence on intermediaries, limited market information, poor visibility of available produce, inefficient communication, weak transaction records and difficulty coordinating buyers and farmers. FarmLink creates an opportunity to use information technology to make agricultural trade more accessible, transparent and efficient.

2. PROBLEM STATEMENT

Many small-scale farmers experience difficulty finding reliable buyers and obtaining timely information about market demand and prices. Buyers, on the other hand, may struggle to identify genuine farmers, available produce, quantities, prices and locations. Traditional communication through physical markets, telephone calls and informal messaging can result in delayed transactions, incomplete records and limited market reach. The absence of a centralized digital platform also makes it difficult for administrators or agricultural organizations to obtain accurate information for planning and decision-making.

FarmLink will address these problems by providing a centralized marketplace and management information system that supports farmer registration, produce listing, buyer registration, product search, ordering, notifications, communication, transaction records and administrative reporting.

3. PROJECT OBJECTIVES

The main objective is to develop an integrated digital marketplace that improves the efficiency and transparency of farm-produce trading. Specific objectives are to:

• Register and manage at least 500 farmers and 1,000 buyers during the initial operational period.

• Allow farmers to create produce listings containing product name, category, photograph, quantity, price, location and availability.

• Enable buyers to search, filter and view produce and submit orders through the platform.

• Provide role-based authentication for farmers, buyers and administrators.

• Maintain accurate records of products, users, orders, payments/statuses and transactions.

• Provide administrative dashboards and reports for monitoring marketplace activities.

• Improve communication between farmers and buyers through system notifications and messaging features.

• Deploy a secure, scalable and user-friendly system that can be expanded to additional agricultural regions.

4. SYSTEM TYPE AND DEVELOPMENT APPROACH

System type: FarmLink is primarily a **Transaction Processing System (TPS)** because it records and processes transactions such as product listings, orders, user registrations and transaction statuses. It also incorporates characteristics of a **Management Information System (MIS)** through dashboards, summaries and reports that support administrative decision-making. The system is also an **e-commerce/agricultural marketplace information system** because it facilitates digital interaction between sellers and buyers.

Development approach: The project will use the **Agile development approach**. Agile is appropriate because marketplace requirements may change after feedback from farmers, buyers and administrators. Development will be divided into manageable iterations. Each iteration will involve requirements refinement, design, implementation, testing and stakeholder feedback. This approach allows important features to be delivered early and improved continuously.

5. PROJECT SCOPE

In scope: user registration and authentication; farmer and buyer profiles; farmer produce management; product photographs; product search and filtering; shopping/order functionality; order tracking; notifications; administrative dashboard; reports; database management; role-based access control; audit records; responsive web interface; testing; deployment and user training.

Out of scope for the initial release: ownership of physical transport vehicles, agricultural production management, automated crop disease diagnosis, international payment settlement, and fully automated delivery. These may be considered in future versions.

6. KEY STAKEHOLDERS

Stakeholder	Interest / Responsibility
Farmers	Register, list produce, manage prices/availability and receive orders.
Buyers	Search produce, place orders and communicate with sellers.
System Administrator	Manage users, products, transactions, reports and security.
Project Sponsor	Provides funding, direction and approval.
Project Team	Analysis, design, development, testing, deployment and maintenance.
Transport/Logistics Partners	Support future delivery and order fulfilment processes.

7. FEASIBILITY ANALYSIS

Economic feasibility: The proposed investment is justified by expected benefits such as increased market reach, reduced marketing costs, improved transaction records, faster communication and potential transaction/service revenue. The initial project budget is estimated at **KSh 2,480,000**, which exceeds the required KSh 1,000,000 threshold. The system is expected to provide measurable value through increased transaction efficiency and market participation.

Operational feasibility: The system is operationally feasible because farmers and buyers can use a web interface to perform activities that are currently performed through manual or fragmented channels. Training, user support and simple navigation will be included to reduce resistance to change.

Technical feasibility: The project can be implemented using established web technologies, a relational database, cloud hosting and secure authentication. The development team will use modern development tools and version control. The system will be designed responsively so it can be accessed from computers and smartphones.

Schedule feasibility: The system is planned for completion within approximately 12 weeks, using iterative Agile development. Major milestones will be reviewed against the schedule to identify delays early.

Legal and contractual feasibility: The system will incorporate appropriate data-protection, privacy, access-control and record-management practices. Users will be informed about how their personal information is collected and used. Relevant Kenyan legal and contractual requirements will be considered before deployment.

Political feasibility: The project requires support from organizational management, agricultural stakeholders and community users. Early stakeholder involvement, transparent decision-making and clear responsibilities will improve acceptance and reduce organizational resistance.

8. MAJOR DELIVERABLES

• Approved requirements specification and project scope document.

• System architecture, database design and user-interface designs.

• Working authentication and role-management module.

• Farmer profile and produce-management module supporting product photographs.

• Buyer marketplace with search, filtering and product details.

• Order-management and transaction-record module.

• Administrator dashboard with management and reporting functions.

• Testing documentation, user acceptance results and defect records.

• Deployed system, user guide and basic administrator training.

• Source code repository and final BPP/report documentation.

9. MILESTONES AND TIMELINE

Milestone	Activities	Duration
1. Initiation & Planning	Problem definition, stakeholders, scope, BPP approval	Week 1

2. Requirements Analysis	Interviews, requirements, process analysis, feasibility	Weeks 2–3
3. System Design	Architecture, database, interfaces and security design	Weeks 4–5
4. Core Development	Authentication, profiles, produce and marketplace modules	Weeks 6–8
5. Integration & Testing	Integration, functional, security and usability testing	Weeks 9–10
6. Deployment & Training	Deployment, documentation and stakeholder training	Week 11
7. Evaluation & Handover	User acceptance, corrections, final reports and handover	Week 12

10. PROJECT BUDGET

Item	Estimated Cost (KSh)
Requirements analysis and consultancy	250,000
Software development and engineering	650,000
UI/UX design and prototyping	180,000
Cloud hosting, domain and infrastructure	180,000
Database, security and backup services	220,000
Testing and quality assurance	200,000
Training, documentation and deployment	180,000
Hardware/networking and operational setup	220,000
Contingency and maintenance reserve	200,000
TOTAL	**2,480,000**

The estimated project budget is KSh 2,480,000, satisfying the assignment requirement that the project budget be greater than KSh 1,000,000.

11. REPORTING AND COMMUNICATION PLAN

Project progress will be communicated to the project sponsor and relevant stakeholders through weekly status updates and milestone reviews. Each weekly report will summarize completed activities, work planned for the following week, achieved deliverables, identified risks, unresolved issues, budget/schedule status and decisions requiring stakeholder attention. Formal milestone reviews will be conducted at the end of requirements analysis, system design, development, testing and deployment.

The project team will use documented meeting minutes, issue logs, task tracking and version control to maintain accountability. Stakeholders will be encouraged to provide feedback during demonstrations so that requirements and system functionality can be adjusted where necessary.

12. EXPECTED OUTCOMES AND SUCCESS MEASURES

• A functional marketplace accessible through common web browsers and mobile devices.

• At least 500 farmers and 1,000 buyers supported during the initial operational period.

• At least 90% of defined functional requirements successfully implemented and tested.

• Product listings can be created, updated, searched and made unavailable when sold or out of stock.

• Orders are recorded with identifiable buyer, seller, product, quantity and status information.

• Administrative reports provide reliable information for operational decision-making.

• Critical security and functional defects are resolved before deployment.

• Stakeholder user acceptance is achieved before formal handover.

13. KEY RISKS AND MITIGATION

Risk	Mitigation
Low user adoption	Conduct demonstrations, training and provide simple interfaces.
Internet/connectivity limitations	Use responsive pages, efficient data transfer and consider future mobile/offline enhancements.
Data security/privacy incidents	Use role-based access, secure authentication, backups, validation and security testing.
Scope creep	Use an approved requirements baseline and formally review change requests.

Budget overruns	Track expenditure against the approved budget and maintain contingency funds.
Delayed development	Use weekly progress reviews, prioritized features and Agile iterations.

14. GITHUB REPOSITORY AND PROJECT HANDOVER

A GitHub repository will be created to maintain the FarmLink source code, documentation, project report and relevant development records. The repository link will be inserted in the final submitted version of this report after the repository has been created. The final project package will include the working system, source code, database scripts/design, documentation and this BPP.

GitHub Repository: [Insert your GitHub repository link here before submission]

15. CONCLUSION

The FarmLink Baseline Project Plan provides a structured roadmap for analyzing, designing, developing, testing and deploying an integrated farm produce marketplace and management information system. The project addresses a practical business problem by improving communication between farmers and buyers, increasing product visibility, strengthening transaction records and supporting management decisions. The proposed Agile approach, measurable deliverables, feasibility analysis, 12-week schedule, reporting framework and KSh 2,480,000 budget provide a realistic baseline for implementation. Any approved changes to scope, schedule, budget or requirements will be documented and reflected in the project baseline.

Prepared for academic assessment – INSY 281: System Analysis & Design

Lecturer: Mr. Jefferson Mwatati | Academic Year: 2026